

Qatar-1b

R-0498

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180802 · created 2026-07-20T14:24:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458332.9012 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.056
Transit depth (Rp/Rs)^2	1.78 +/- 1.49 [%]
Orbital Inclination (inc)	84.67 +/- 2.9
Airmass coefficient 1 (a1)	1.4 +/- 1.04
Airmass coefficient 2 (a2)	-0.39 +/- 1.28
Scatter in the residuals of the lightcurve fit is	81.81 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	6.76
Transit Duration (day)	0.065 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.134 $\pm$ 0.056 vs 0.1463 (deviation 0.22 σ)
Duration fit vs published	0.065 d vs 0.0692 d (deviation 0.21 σ)
Mid-time O-C	2.8 min
Depth SNR	2.4
Residual scatter	81.81 %

Light curve

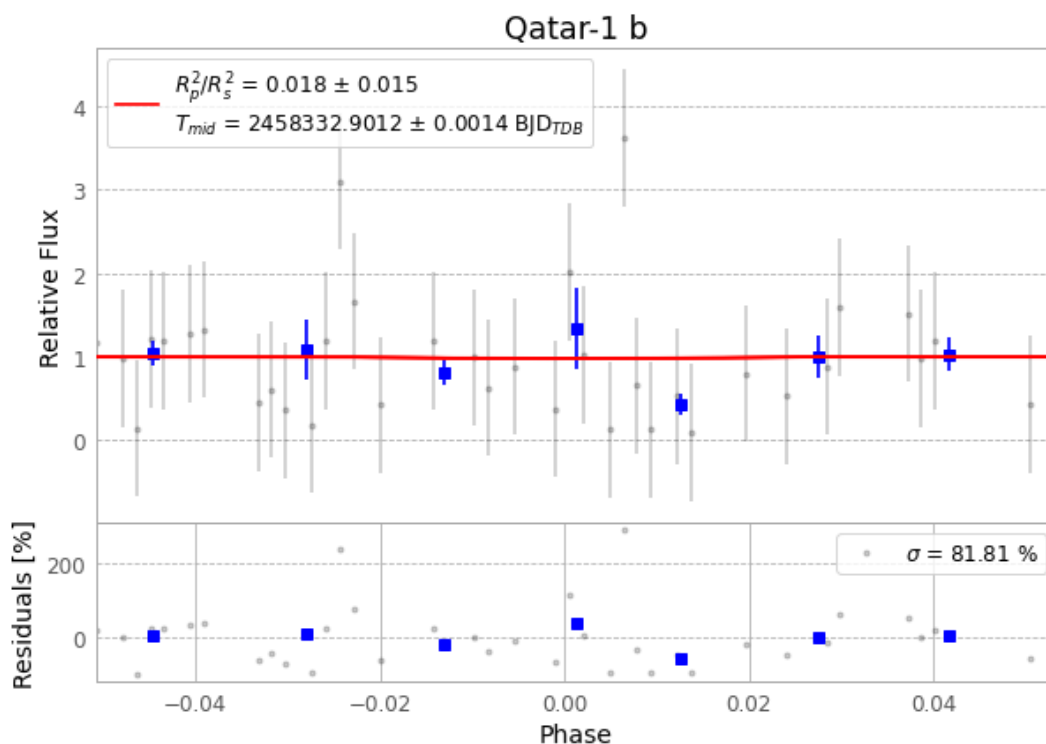


Figure 1 — FinalLightCurve_Qatar-1 b_2018-08-02.png (SHA-256 adec933c329cbafe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 98], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 78ce7afabdb42e93...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

73 FITS frames (48 MB), Qatar-1180802074812.FITS → Qatar-1180802112412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180802.log (ce124d95e087...), FinalLightCurve_Qatar-1 b_2018-08-02.png (adec933c329c...), FinalLightCurve_Qatar-1 b_2018-08-02.csv (93f3264d1ace...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 14:24Z.